

MCTA 3371 Computational Intelligence: Mini Project



Intelligent Mobile Robot Navigation Using Hybrid Fuzzy Logic and Genetic Algorithm

Section 2

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Abstract

This project presents an intelligent mobile robot navigation system using a Hybrid Fuzzy Logic and Genetic Algorithm (Fuzzy-GA) approach. The controller was developed and tested in MATLAB using two different maze environments: a simple map and a complex map. Fuzzy Logic was used to make navigation decisions based on environmental information, while the Genetic Algorithm helped improve path selection and navigation efficiency. The robot was required to move from a starting point to a goal point while avoiding obstacles. Simulation results showed that the robot successfully reached the target in both environments without collisions. The proposed Hybrid Fuzzy-GA controller demonstrated effective obstacle avoidance, adaptive decision-making, and reliable navigation performance in different maze complexities.

1.Introduction

Mobile robot navigation is an important area in computational intelligence, where a robot must move from a start point to a goal while avoiding obstacles. Traditional navigation methods often require complex mathematical models, making them less suitable for uncertain environments. Soft computing techniques such as Fuzzy Logic (FL), Genetic Algorithms (GA), and hybrid approaches provide flexible and intelligent solutions for navigation tasks.

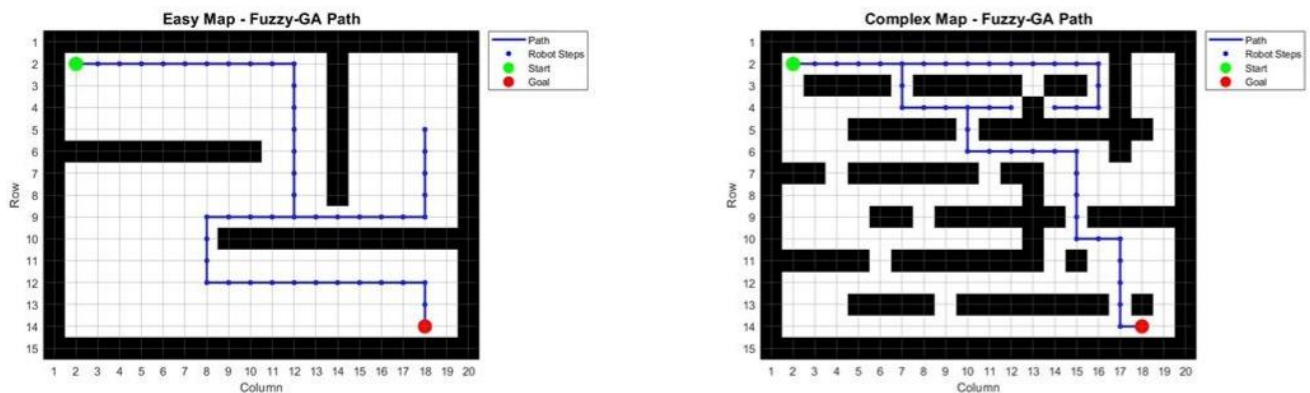
In this project, a Hybrid Fuzzy Logic–Genetic Algorithm (Fuzzy-GA) controller was developed to navigate a mobile robot in a 2D maze environment. The robot was tested on both simple and complex maps to evaluate its ability to reach the goal while avoiding obstacles. By combining fuzzy decision-making with genetic optimization, the controller was able to select efficient paths and adapt to different maze

configurations. The performance of the system was assessed based on successful navigation and obstacle avoidance.

2. Objectives

- Design an intelligent navigation controller using computational intelligence techniques.
- Simulate robot navigation in different environments.
- Compare navigation performance using hybrid Fuzzy-GA methods.
- Evaluate success rate, path efficiency, obstacle avoidance and navigation time.

3. Environment Setup and Robot Modelling



Environment Setup

- A 2D grid-based maze environment was developed in MATLAB.
- Two maps were created:
 - **Easy Map** – fewer obstacles and wider paths.
 - **Complex Map** – more obstacles and narrower paths.

- Each map contains:
 - Start Point (Green)
 - Goal Point (Red)
 - Obstacles/Walls (Black)
 - Free Space (White)

Robot Modelling

- Robot is represented as a **point robot** in a grid environment.
- The robot moves in **discrete steps**.
- Available movements:
 - Up
 - Down
 - Left
 - Right
- Collision detection prevents movement into obstacles or outside the map boundaries.

Controller Inputs

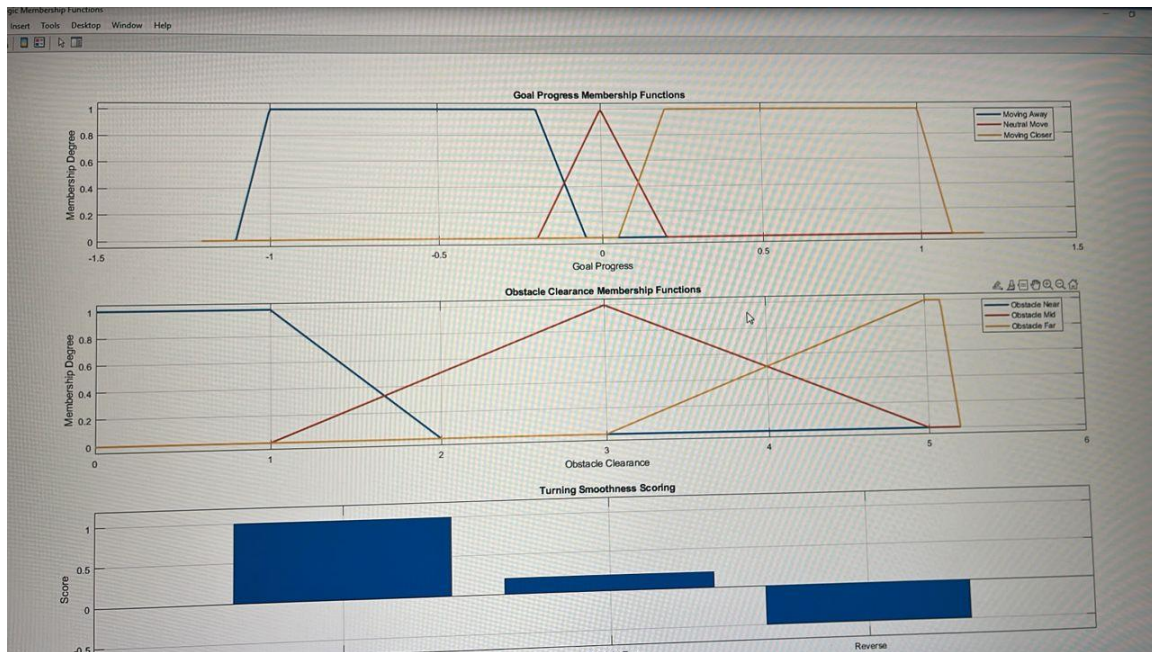
- Distance to obstacles
- Distance to goal
- Available free path in each direction

Controller Outputs

- Next movement direction
- Optimal path toward the goal

Navigation Objective

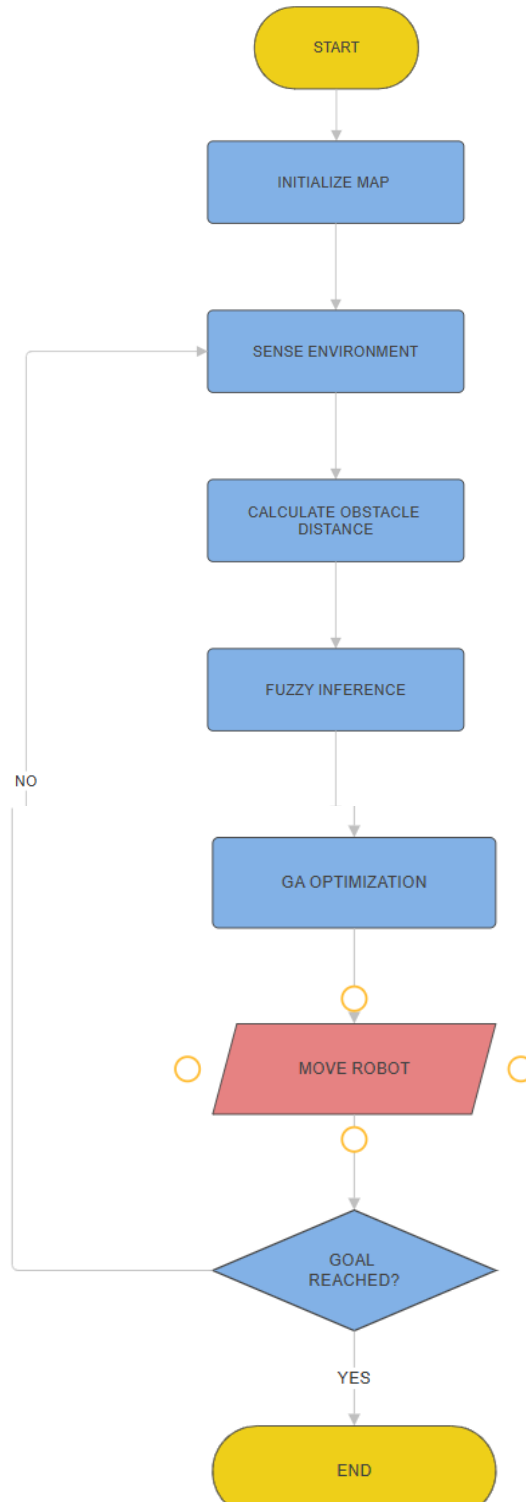
- Reach the goal successfully.
- Avoid collisions with obstacles.
- Minimize path length and navigation time.



4. Methodology

The methodology of this project involves designing and simulating an intelligent mobile robot navigation system using a Hybrid Fuzzy Logic and Genetic Algorithm (Fuzzy-GA) approach. First, two 2D maze environments, namely an easy map and a complex map, were created in MATLAB with predefined start and goal positions as well as obstacles. The robot was modelled as a point robot that moves in discrete steps within the grid environment. During navigation, the robot continuously senses its surroundings by evaluating the distance to obstacles and the direction of the goal. This information is processed by the fuzzy logic controller, which applies predefined membership functions and fuzzy rules to determine the most suitable movement direction. The Genetic Algorithm is then used to optimize the navigation process by improving path selection and reducing inefficient movements. The robot repeatedly updates its position based on the controller's decision until it successfully reaches the goal while avoiding collisions. Finally, the navigation performance was evaluated by observing the robot's ability to reach the destination, avoid obstacles, and generate an efficient path in both maze environments.

5. Flowchart



6. Simulation Results

- **6.1 Easy Map Result**

The robot successfully navigated from the start position to the goal position without colliding with obstacles.

Observations:

- Smooth navigation.
- Few direction changes.
- Efficient path planning.
- Goal reached successfully.

- **6.2 Complex Map Result**

The robot successfully reached the goal despite the presence of multiple obstacles and narrow passages.

Observations:

- Increased path complexity.
- More turning actions required.
- Longer navigation path.
- Successful obstacle avoidance.

The simulation results demonstrate that the Hybrid Fuzzy-GA controller can effectively handle both simple and complex navigation environments.

7. Data Collection

Data were collected during each simulation run by recording the robot's movement from the start position to the goal position. The collected data included total path length travelled, number of movement steps, successful goal completion, obstacle avoidance performance, and occurrence of invalid moves. The robot's path coordinates were stored and visualized to evaluate navigation behavior. Data were obtained for both the easy and complex maps to allow comparison of controller performance under different environmental conditions.

8. Data Analysis

The collected results show that the Hybrid Fuzzy-GA controller successfully guided the robot to the goal in both environments without collisions. The easy map produced a shorter path and required fewer movement steps due to its simpler structure and lower obstacle density. In contrast, the complex map required additional turns and a longer travel distance because of the increased number of obstacles and maze constraints. Despite the increased difficulty, the controller maintained successful navigation performance, demonstrating effective decision-making, obstacle avoidance capability, and adaptability to different map complexities.

9. Discussion

The Hybrid Fuzzy-GA controller demonstrated effective navigation performance across different maze environments. Fuzzy Logic enabled intelligent decision-making under uncertain conditions, while the Genetic Algorithm improved path optimization. The combination of both

techniques resulted in reliable navigation and obstacle avoidance. The controller successfully adapted to varying maze complexities without requiring detailed mathematical modelling of the robot.

10. Conclusion

This project successfully developed an intelligent mobile robot navigation system using a Hybrid Fuzzy Logic and Genetic Algorithm approach. The robot was able to navigate through both simple and complex maze environments while avoiding obstacles and reaching the target location. The simulation results demonstrate that hybrid computational intelligence techniques can provide effective solutions for autonomous navigation problems. Future improvements may include dynamic obstacle avoidance, 3D environments, and integration with real-world robotic platforms.

11. AI Usage Appendix

AI Tools Used

- ChatGPT

Example Prompts

1. "Generate MATLAB code for a Hybrid Fuzzy-GA maze navigation system."
2. "Explain fuzzy membership functions for robot navigation."
3. "Help write project report sections for intelligent robot navigation."

Usage of AI

The AI tool was used to assist in code explanation, report writing, and understanding computational intelligence concepts. All generated content was reviewed, modified, and verified by the group members before inclusion in the final project.

Responsibility Statement

The group members take full responsibility for the final implementation, simulation results, report contents, and conclusions presented in this project.

Appendix B – Individual Contributions

Member 1: Coding and implementation.

Member 2: Simulation and testing.

Member 3: Documentation and presentation.

(Modify according to actual contributions.)

12. References

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[3] Kumar, A., Sahasrabudhe, A., & Nirgude, S. (2024). *Fuzzy Logic Control for Indoor Navigation of Mobile Robots*. arXiv.
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